

A simple cure for numerical shock instability in the HLLC Riemann solver

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ARTICLE INFO

Article history:

Received 4 February 2018

Received in revised form 10 November 2018

Accepted 12 November 2018

Available online 15 November 2018

Keywords:

Carbuncle phenomenon

Numerical shock instability

Riemann solver

Shock stable HLLC scheme

Contact and shear preserving ability

Stability analysis

ABSTRACT

The Harten–Lax–van Leer with contact (HLLC) approximate Riemann solver is known to be plagued by various forms of numerical shock instability. In this paper, we propose a new framework for developing shock stable versions of the HLLC scheme for the Euler system of equations without compromising on its contact and shear preserving ability. The proposed framework, termed as **Selective Wave Modification (SWM)**, identifies and enhances the magnitude of the inherent HLL-type diffusive component within the HLLC scheme in the vicinity of a normal shock front. This is achieved by suitably increasing the magnitudes of the nonlinear wave speed estimates appearing in them through a dissipation parameter ϵ . A linear perturbation analysis is performed to gauge the effectiveness of the proposed framework in damping unwanted perturbations in physical quantities and is also used to estimate a von-Neumann-type stability bound on the CFL number associated with its use. Two distinct methods to estimate ϵ , which results in HLLC-SWM-E (Eigenvalue based) and HLLC-SWM-P (Pressure based) schemes, are proposed and compared. Further, a matrix based stability analysis is used to show that the proposed schemes can be configured to remain shock stable over a wide range of free stream Mach numbers. Numerical results confirm that the proposed schemes are capable of computing shock stable solutions for a wide variety of problems. On viscous flows, the HLLC-SWM-P variant is found to be the most accurate. Performance of these schemes suggests that it is possible to construct shock stable upwind schemes that simultaneously retains accuracy on the linearly degenerate wavefields.

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1. Introduction

Accurate and robust computation of high-speed gas dynamical flows continues to be a challenge due to the presence of strong shocks, expansion fans, contact surfaces and their mutual interactions. The approximate Riemann solvers are one of the most popular methods available for numerical computations of such flows. However, it is well known that the contact and shear preserving variants belonging to this class like the Roe scheme [1], the HLLC scheme [2] and the HLEM scheme [3] are susceptible to numerical shock instability phenomenon during multidimensional computations involving strong normal shocks while, on the other hand, the contact and shear diffusive schemes like the HLL scheme [4] and the HLE scheme [5] are free from them. These spurious solutions are characterized by rapid amplification of benign perturbations occurring during computations. Most commonly, the instability manifest as oscillations in the shock profile, polluted aftershock val-

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ues, growth in error norms, erroneous jumps in quantities across the shock and in extreme cases complete breakup of the shock front itself. Additionally, the occurrence of the instability is also highly sensitive to parameters like grid aspect ratio [6], inflow Mach number [7], numerical shock structure [7,8], perturbations in upstream supersonic flow [7] and order of accuracy of the solution [9] among other things. A detailed catalogue of these failures can be found in [7,10,11].

Several attempts have been made to discern the exact cause of these spurious solutions. Gressier et al. [9] conjectured that the notions of strict shock stability and accuracy on linearly degenerate wavefields are incompatible for upwind schemes. Quirk [10] conjectured that instability is caused because of coupling between pressure perturbations and density perturbations along the length of the shock which affects the numerical shock speed. Kun Xu [12] and others [7,8,13] located the source of instability to be within the numerical shock region. Liou [14] clarified that for a scheme to remain shock stable, it must not contain pressure terms in its mass flux discretization. He argued that presence of such terms will promote unphysical interactions between the pressure fields and density fields and result in instability. Some researchers [13,15,16] identify that inadequate dissipation of shear wave could result in the instability. Sanders et al. [17] analyzed the numerical truncation error equation to establish that shock instability in upwind schemes arise due to lack of cross-flow dissipation provided by these schemes within a numerical shock region. Shen et al. [18] showed how the Roe scheme suffers an overall loss of numerical dissipation along a numerically perturbed shock front due to vanishing of terms corresponding to the linearly degenerate wavefields within the dissipation matrix. The reduced numerical dissipation, which is incapable of attenuating perturbations in physical quantities, is thought to cause existence of unphysical variation of the conserved quantity ρu along the shock front and result in the instability. Xie et al. [13] showed, in the context of HLL-based schemes, that unphysical variation of the quantity ρu in the vicinity of the shock front is due to perturbation errors within the numerical shock region which is amplified over time by a contact and shear preserving scheme. Dumbser et al. [7] developed a useful matrix based stability analysis and showed that the shock instability is a manifestation of an inherent instability associated with a numerical scheme computing a normal shock front.

Based on these observations, various methods have been proposed in the literature to cure approximate Riemann solvers of this deficiency. All of these cures are based on introducing a certain amount of numerical dissipation to counter the instability by attenuating unwanted perturbations in flow variables. Thus, there are hybrid Riemann solvers like Roe-HLLE [19–21] and HLLC-HLL [15,22–24] and rotated Riemann solvers [16,21,25] that treat the instability phenomenon with varying levels of success. While the hybrid methods require explicit switching sensors and careful choice of complementary Riemann solvers, the rotated methods are computationally at least twice as expensive to implement as their grid-aligned counterparts. As yet, one of the simplest methods to deal with the shock instability in Roe scheme involves increasing its overall numerical dissipation by modifying the eigenvalues of its linearized Jacobian matrix; including the ones corresponding to the linearly degenerate waves. Peery and Imlay [26] demonstrated a cure for ‘Carbuncle phenomenon’ for the Roe scheme based on this idea. The modification they proposed used a direction based pressure gradient to tune the magnitude of the numerical dissipation introduced. Lin [27] pointed out the limitations of this pressure-based sensor and proposed an improved modification that offered better accuracy on viscous flows. Sanders et al. [17] suggested using the idea of an entropy fix on the linearly degenerate waves in conjunction with a simple ‘H’-shaped multidimensional stencil to infuse the requisite amount of numerical dissipation to cure the Roe scheme. Later, Pandolfi et al. [11] pointed out a more efficient way to implement the same that avoids unwanted dissipation at shear layers for anisotropic grids. However, as noted by some authors [7,9], there is no physical justification for modifying the eigenvalues belonging to the linearly degenerate waves using an entropy fix.

Incidentally, majority of cures for shock instability have been developed for the Roe scheme. However, Roe scheme suffers several drawbacks like requirement of an entropy fix on its nonlinear wave speeds to admit entropy satisfying solutions [28], demand for knowledge of the full eigenstructure of the governing equations, difficulty in being easily extendable to general system of hyperbolic equations with complex equations of state [29] and lack of positivity in low density flows [3]. These shortcomings of the Roe scheme has brought forth an increasing interest in another equally accurate and efficient contact and shear preserving Riemann solver namely, the HLLC Riemann solver [2]. Unfortunately, the HLLC scheme also succumbs to numerical shock instability. Hence the need for preventing the occurrence of these spurious solutions in the HLLC scheme is paramount.

In this paper, we present a very simple and inexpensive strategy to develop shock stable versions of the HLLC scheme. The strategy is termed **Selective Wave Modification (SWM)** and the HLLC scheme implementing it is referred to as **HLLC-SWM**. The novelty of the formulation lies in first decomposing the HLLC scheme into a diffusive HLL component and an antidiffusive component that recovers the accuracy on linearly degenerate waves. Once distinguished, we propose a strategy to adaptively enhance the magnitude of numerical dissipation offered by this diffusive HLL component in the vicinity of shocks, by increasing the magnitudes of the nonlinear wave speed estimates appearing in them through a dissipation parameter. In contrast to the cures proposed in [11,17,27], the present method only targets certain nonlinear wave speeds that reside in the diffusive HLL component within the HLLC scheme, while leaving the middle wave speed which is crucial to the contact and shear preserving ability of the scheme completely unmodified. Various theoretical analyses and an array of numerical test problems confirm that this simple technique is found to provide sufficient numerical dissipation in the vicinity of the shock front to eliminate shock instability.

The outline of the paper is as follows. In Sections 2 and 3, a brief outline of the governing equation and the Finite Volume framework used in the present work are discussed. Section 4 reviews the HLL and the HLLC Riemann solvers. In Section 5, we discuss the formulation of the HLLC-SWM framework and prove the effectiveness of the proposed strategy in

increasing the overall numerical dissipation of the HLLC scheme. In Section 6, a linear perturbation analysis is performed to demonstrate how the HLLC-SWM framework is effective in suppressing perturbations in primitive variables. In Sections 7 and 8, we describe two approaches for practical implementation of the HLLC-SWM framework that assures shock stability over a wide range of free stream Mach numbers. In Section 9, these implementations of the HLLC-SWM framework are extensively tested on a series of numerical test problems. Some concluding remarks are given in Section 10.

2. Governing equations

The governing equations for the two-dimensional inviscid compressible flow can be expressed in their conservative form as,

$$\frac{\partial \hat{\mathbf{U}}}{\partial t} + \frac{\partial \hat{\mathbf{F}}(\hat{\mathbf{U}})}{\partial x} + \frac{\partial \hat{\mathbf{G}}(\hat{\mathbf{U}})}{\partial y} = 0 \quad (1)$$

where, $\hat{\mathbf{U}}, \hat{\mathbf{F}}(\hat{\mathbf{U}}), \hat{\mathbf{G}}(\hat{\mathbf{U}})$ are the vector of conserved variables and x - and y -directional fluxes respectively. These are given by,

$$\hat{\mathbf{U}} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \end{bmatrix}, \hat{\mathbf{F}}(\hat{\mathbf{U}}) = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ (\rho E + p)u \end{bmatrix}, \hat{\mathbf{G}}(\hat{\mathbf{U}}) = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ (\rho E + p)v \end{bmatrix} \quad (2)$$

In the above expressions, ρ, u, v, p and E stands for density, x -directional and y -directional velocities in the global coordinates, pressure and specific total energy. The system of equations are closed through the equation of state,

$$p = (\gamma - 1) \left(\rho E - \frac{1}{2} \rho (u^2 + v^2) \right) \quad (3)$$

where, γ is the ratio of specific heats. Present work assumes a calorically perfect gas with $\gamma = 1.4$. A particularly useful form of the Eq. (1) is the integral form given by,

$$\frac{d}{dt} \int_{\Omega} \hat{\mathbf{U}} d\Omega + \oint_{\partial\Omega} [(\hat{\mathbf{F}}, \hat{\mathbf{G}}) \cdot \mathbf{n}] dS = 0 \quad (4)$$

where, Ω denotes a control volume (or an area in the case of two-dimensions) which represents an arbitrary finite region of the flow, fixed in space and bounded by the closed surface $\partial\Omega$. The symbol dS denotes an infinitesimally small surface element on $d\Omega$. The symbol $\mathbf{n}$ denotes the outward pointing unit normal vector to the surface $d\Omega$.

3. Finite volume discretization

In this paper we seek a Finite Volume numerical solution of Eq. (4) by discretizing the equation on a computational mesh consisting of structured quadrilateral cells. For a typical cell i belonging to this mesh, a semi-discretized version of such a solution can be written as,

$$\frac{d\hat{\mathbf{U}}_i}{dt} = -\frac{1}{|\Omega|_i} \sum_{k=1}^4 [(\hat{\mathbf{F}}, \hat{\mathbf{G}})_k \cdot \mathbf{n}_k] \Delta s_k \quad (5)$$

where, $\hat{\mathbf{U}}_i$ is an appropriate cell-averaged conserved state vector, $(\hat{\mathbf{F}}, \hat{\mathbf{G}})_k$ denotes appropriate time-averaged global x - and y -directional flux vectors at the mid-point of each interface I_k while, $\mathbf{n}_k$ and Δs_k denotes the unit normal vector and the length of each I_k interface respectively. Throughout this paper, the interface flux $(\hat{\mathbf{F}}, \hat{\mathbf{G}})_k \cdot \mathbf{n}_k$ are obtained by employing an approximate Riemann solver. A conventional two-state approximate Riemann solver uses the rotational invariance property of Euler equations to express the term $(\hat{\mathbf{F}}, \hat{\mathbf{G}})_k \cdot \mathbf{n}_k$ as,

$$\frac{d\hat{\mathbf{U}}_i}{dt} = -\frac{1}{|\Omega|_i} \sum_{k=1}^4 [\mathbf{T}_k^{-1} \mathbf{F}(\mathbf{U}_L, \mathbf{U}_R)] \Delta s_k \quad (6)$$

where, $\mathbf{U}_L = \mathbf{T}_k(\mathbf{U}_i)$, $\mathbf{U}_R = \mathbf{T}_k(\mathbf{U}_m)$ indicates the initial conditions of a local Riemann problem across k th interface shared by the cells i and m and $\mathbf{F}(\mathbf{U}_L, \mathbf{U}_R)$ denote the time-averaged, local, two-state Riemann flux at that interface. The matrices $\mathbf{T}_k$ and $\mathbf{T}_k^{-1}$ are rotation matrices at the k th interface and are described in [30].

In the next section we briefly describe two approximate Riemann solvers namely the HLL scheme and the HLLC scheme that can be used to estimate the flux $\mathbf{F}(\mathbf{U}_L, \mathbf{U}_R)$ at a given interface. To avoid extra notations, henceforth $\mathbf{F}$ simply represent the local Riemann flux at any interface with outward pointing normal $\mathbf{n}$.

4. A review of two classic Riemann solvers

The following subsections brief two classic Riemann solvers on which the present work builds upon.

4.1. The HLL Riemann solver

The HLL Riemann solver, devised by Harten, Lax and van Leer [4] assumes a wave structure consisting of two waves that separates three constant states: the left and right states of the local Riemann problem and an average Riemann solution state enclosed between them. The two waves, represent two genuinely nonlinear waves like shocks or expansion fans that arise as a solution of the local Riemann problem at an interface. Using this wave structure, the average Riemann solution state $\mathbf{U}_{HLL}^*$ can be derived through application of integral form of the conservation laws as,

$$\mathbf{U}_{HLL}^* = \frac{S_R \mathbf{U}_R - S_L \mathbf{U}_L + \mathbf{F}_L - \mathbf{F}_R}{S_R - S_L} \quad (7)$$

Using Rankine–Hugoniot conditions across these waves, the HLL Riemann flux can then be written as,

$$\mathbf{F}_{HLL} = \frac{S_R \mathbf{F}_L - S_L \mathbf{F}_R + S_L S_R (\mathbf{U}_R - \mathbf{U}_L)}{S_R - S_L} \quad (8)$$

Here S_L and S_R are numerical approximations to the speeds of the left most and right most running characteristics that emerge as the solution of the local Riemann problem at the cell interface of interest. In this work, S_L and S_R are computed as [5],

$$\begin{aligned} S_L &= \min(0, u_{nL} - a_L, \tilde{u}_n - \tilde{a}) \\ S_R &= \max(0, u_{nR} + a_R, \tilde{u}_n + \tilde{a}) \end{aligned} \quad (9)$$

where, $u_{nL,R}$ are the normal velocities across an interface, $a_{L,R}$ are the respective sonic speeds of the left and right states and $\tilde{u}_n, \tilde{a}$ are the standard Roe-averaged quantities at the interface [1]. Using these wave speed estimates, the HLL scheme is also known as the HLLC scheme [5]. It has been shown that using these wavespeeds, the HLLC scheme is both positivity preserving and entropy satisfying [5]. Although the HLLC scheme is very robust for shock capturing applications, one of its major drawback is its inability to resolve the linearly degenerate waves of the Euler system that govern the contact and the shear phenomenon. Such an inability severely restricts the scheme from being used in a full Navier–Stokes solver because capability to resolve these waves is generally regarded as a prerequisite for accurate resolution of viscous phenomenon [31,32]. However, Gressier et al. [9] have conjectured that upwind schemes that possess exact contact ability cannot be free from numerical shock instability. Hence by corollary, the shock stable behavior of the HLLC scheme is attributed to its diffusive nature [11].

4.2. The HLLC Riemann solver

Toro et al. [2] mitigated the shortcoming of the HLL Riemann solver by introducing the contact and shear preserving feature into it. The resulting solver is called the HLLC (C for Contact) scheme. This improvement was achieved by adding a third wave, called a contact wave with speed S_M , to the two-wave HLL wave structure. Once again, by using the integral form of the conservation laws and appropriate flow conditions relating to the additional wave, closed form expressions for the average Riemann solution states $\mathbf{U}_{*L/R}^{HLLC}$ that exist across the additional wave S_M can be obtained as,

$$\mathbf{U}_{*L/R}^{HLLC} = \rho_{L/R} \left(\frac{S_{L/R} - u_{nL/R}}{S_{L/R} - S_M} \right) \begin{pmatrix} 1 \\ S_M \\ u_{tL/R} \\ \left(\frac{(\rho E)_{L/R}}{\rho_{L/R}} + (S_M - u_{nL/R})(S_M + \frac{p_{L/R}}{\rho_{L/R}(S_{L/R} - u_{nL/R})}) \right) \end{pmatrix} \quad (10)$$

where, $u_{tL/R}$ denote the local tangential velocity at an interface. Based on these, the HLLC interface flux $\mathbf{F}_{HLLC}$ can be written as,

$$\mathbf{F}_{HLLC} = \begin{cases} \mathbf{F}_L + S_L (\mathbf{U}_{*L}^{HLLC} - \mathbf{U}_L), & \text{if } S_L \leq 0 \leq S_M \\ \mathbf{F}_R + S_R (\mathbf{U}_{*R}^{HLLC} - \mathbf{U}_R), & \text{if } S_M \leq 0 \leq S_R \end{cases} \quad (11)$$

In the above expressions, S_L and S_R can be obtained using Eq. (9). Batten et al. [33] provides a closed form expression for S_M as,

$$S_M = \frac{p_R - p_L + \rho_L u_L (S_L - u_L) - \rho_R u_R (S_R - u_R)}{\rho_L (S_L - u_L) - \rho_R (S_R - u_R)} \quad (12)$$

The HLLC Riemann solver is one of the simplest known Riemann solver to be able to resolve all the waves of the Euler system accurately. Like the HLL Riemann solver, it is also an entropy satisfying and positively conservative scheme under appropriate wave speed selection [2,33]. However, regardless of all the attractive features the scheme possesses, the HLLC Riemann solver is known to be highly susceptible to numerical shock instability. In the next section, a simple and inexpensive method of protecting the HLLC scheme from the instability phenomenon is presented. The idea involves first splitting the standard HLLC scheme into a HLL-type diffusive component and an antidiffusive component responsible for restoring accuracy on the linearly degenerate wavefields. Then, a multidimensional, solution dependent strategy is used to enhance the magnitude of this diffusive component in the vicinity of shocks by increasing the magnitudes of the nonlinear wave speed estimates appearing in them. The additional numerical dissipation made available through this strategy is found to prevent the occurrence of shock instability in this scheme.

5. Formulation of a shock stable HLLC scheme

We propose the following strategy to save the HLLC scheme from shock instability. We first decompose the HLLC scheme into an HLL-type diffusive component and an antidiffusive component responsible for restoration of accuracy on contact and shear waves. To achieve this, we propose to rewrite the HLLC scheme as,

$$\mathbf{F}_{HLLC} = \begin{cases} \mathbf{F}_{HLL} + \mathbf{A}_{L,HLLC}, & \text{if } S_L \leq 0 \leq S_M \\ \mathbf{F}_{HLL} + \mathbf{A}_{R,HLLC}, & \text{if } S_M \leq 0 \leq S_R \end{cases} \quad (13)$$

where, $\mathbf{F}_{HLL} = \mathbf{F}_{L/R} + S_{L/R}(\mathbf{U}_{HLL}^* - \mathbf{U}_{L/R})$ represents the HLL-type diffusive component embedded within the HLLC scheme and $\mathbf{A}_{L/R,HLLC} = S_{L/R}(\mathbf{U}_{L/R}^{*HLLC} - \mathbf{U}^{*HLL})$ represents the antidiffusive component. The diffusive component $\mathbf{F}_{HLL}$ can be further recast into a combination of central terms plus upwind dissipation as,

$$\mathbf{F}_{HLL} = \frac{1}{2}(\mathbf{F}_L + \mathbf{F}_R) + \mathbf{D}_{HLL} \quad (14)$$

where, $\mathbf{D}_{HLL}$ denotes the upwind numerical dissipation term offered by the diffusive component and can be written down as [34],

$$\mathbf{D}_{HLL} = \frac{S_R + S_L}{2(S_R - S_L)}(\mathbf{F}_L - \mathbf{F}_R) - \frac{S_L S_R}{(S_R - S_L)}(\mathbf{U}_L - \mathbf{U}_R) \quad (15)$$

We note that $\mathbf{D}_{HLL}$ can also be written in an equivalent form inspired from [35] as,

$$\mathbf{D}_{HLL} = \frac{|S_R| - |S_L|}{2(S_R - S_L)}(\mathbf{F}_L - \mathbf{F}_R) + \frac{|S_L|S_R - |S_R|S_L}{2(S_R - S_L)}(\mathbf{U}_L - \mathbf{U}_R) \quad (16)$$

It is easy to notice that the overall numerical dissipation $\mathbf{D}_{HLL}$ provided by the HLLC scheme appears as a part of the HLL-type diffusive component, while the accuracy on contact and shear waves, that makes the HLLC scheme more accurate than the HLL scheme, is provided through the antidiffusive components $\mathbf{A}_{L/R,HLLC}$. Once the inherent numerical dissipation term $\mathbf{D}_{HLL}$ is identified, a strategy to increase its magnitude is sought near shock waves. Davis [36] has noted that the magnitude of the nonlinear wavespeed estimates S_L and S_R can directly affect the amount of numerical dissipation provided by the HLL scheme. Hence, it is envisioned that additional numerical dissipation to deal with the shock instability phenomenon can be introduced into the HLLC scheme through careful manipulations of the wave speed estimates S_L and S_R appearing in $\mathbf{D}_{HLL}$. In the present work, a modified numerical dissipation term $\bar{\mathbf{D}}_{HLL}$ is proposed,

$$\bar{\mathbf{D}}_{HLL} = \frac{|\bar{S}_R| - |\bar{S}_L|}{2(\bar{S}_R - \bar{S}_L)}(\mathbf{F}_L - \mathbf{F}_R) + \frac{|\bar{S}_L|S_R - |\bar{S}_R|S_L}{2(\bar{S}_R - \bar{S}_L)}(\mathbf{U}_L - \mathbf{U}_R) \quad (17)$$

where,

$$\begin{aligned} \bar{S}_L &= S_L - \alpha \epsilon^L \\ \bar{S}_R &= S_R + \alpha \epsilon^R \end{aligned} \quad (18)$$

In the above expressions, the term $\epsilon^{L,R}$ is an appropriately chosen dissipation parameter that increases the magnitude of the leftmost and rightmost running nonlinear wave speed estimates S_L , S_R respectively, which appears in the solution of the local Riemann problem at any interface. The parameter α is a coefficient through which the dissipation parameter $\epsilon^{L,R}$ introduced into S_L , S_R can be controlled. Both $\epsilon^{L,R}$ and α are by definition nonnegative quantities and will be defined completely later.

It has been recently demonstrated that only the first and third Riemann flux components of the HLLC scheme, especially on the transverse interfaces adjoining a numerical shock front, plays a critical role in triggering shock instability [18,37].

Hence, we choose to employ the modified numerical dissipation term $\bar{\mathbf{D}}_{HLL}$ only on these flux components while retaining the original numerical dissipation term $\mathbf{D}_{HLL}$ on the other flux components. Based on the above discussions, the flux expression for the HLLC-SWM scheme can be written as,

If $S_L \leq 0 \leq S_M$,

$$\mathbf{F}_{HLLC-SWM} = \begin{cases} \left\{ \frac{1}{2}(\mathbf{F}_L + \mathbf{F}_R) + \bar{\mathbf{D}}_{HLL} + \mathbf{A}_{L,HLLC} \right\}, & \text{if 1st and 3rd Riemann flux components} \\ \left\{ \frac{1}{2}(\mathbf{F}_L + \mathbf{F}_R) + \mathbf{D}_{HLL} + \mathbf{A}_{L,HLLC} \right\}, & \text{if 2nd and 4th Riemann flux components} \end{cases} \quad (19)$$

Else if $S_M \leq 0 \leq S_R$,

$$\mathbf{F}_{HLLC-SWM} = \begin{cases} \left\{ \frac{1}{2}(\mathbf{F}_L + \mathbf{F}_R) + \bar{\mathbf{D}}_{HLL} + \mathbf{A}_{R,HLLC} \right\}, & \text{if 1st and 3rd Riemann flux components} \\ \left\{ \frac{1}{2}(\mathbf{F}_L + \mathbf{F}_R) + \mathbf{D}_{HLL} + \mathbf{A}_{R,HLLC} \right\}, & \text{if 2nd and 4th Riemann flux components} \end{cases} \quad (20)$$

Note that the contact wave speed S_M and the wave speeds S_L, S_R , associated with the antidiffusive component, are left completely unaffected by this modification. This is done purposefully to avoid affecting the contact and shear capturing ability of the HLLC scheme. Owing to this *selective wave modification* (SWM) strategy, the proposed modified version of the HLLC scheme will be henceforth referred to in this work as HLLC-SWM scheme.

5.1. Effect of $\epsilon^{L,R}$ on numerical dissipation of the HLLC scheme

In this section, we aim to analytically demonstrate how the $\epsilon^{L,R}$ parameter, introduced by the modified numerical dissipation term $\bar{\mathbf{D}}_{HLL}$, results in an overall increase of the numerical dissipation provided by the HLLC scheme. Although the explanation is provided through the expression of $\mathbf{F}_{HLLC-SWM}$ corresponding to the condition $S_L \leq 0 \leq S_M$, it remains symmetrically valid for the case when $S_M \leq 0 \leq S_R$.

Let a_0 and a_1 be coefficients associated with the unmodified numerical dissipation term $\mathbf{D}_{HLL}$ given in Eq. (16) and defined as,

$$\begin{aligned} a_0 &= \frac{|S_R| - |S_L|}{2(S_R - S_L)} \\ a_1 &= \frac{|S_L|S_R - |S_R|S_L}{2(S_R - S_L)} \end{aligned} \quad (21)$$

Let $\bar{a}_0$ and $\bar{a}_1$ be the corresponding coefficients associated with the modified numerical dissipation term $\bar{\mathbf{D}}_{HLL}$ given in Eq. (17) and defined as,

$$\begin{aligned} \bar{a}_0 &= \frac{|S_R + \epsilon^R| - |S_L - \epsilon^L|}{2(S_R - S_L)} \\ \bar{a}_1 &= \frac{|S_L - \epsilon^L|S_R - |S_R + \epsilon^R|S_L}{2(S_R - S_L)} \end{aligned} \quad (22)$$

For the sake of convenience, in the above expressions, the value of α is taken to be 1. Since in subsonic flows, $S_L < 0$ and $S_R > 0$, it becomes immediately evident that the denominator of $\bar{a}_0$ and $\bar{a}_1$ will be always positive. Further, the coefficients in Eq. (22) can be simplified as,

$$\begin{aligned} \bar{a}_0 &= \frac{|S_R| - |S_L|}{2(S_R - S_L)} + \frac{\epsilon^R - \epsilon^L}{2(S_R - S_L)} \\ \bar{a}_1 &= \frac{|S_L|S_R - |S_R|S_L}{2(S_R - S_L)} + \frac{\epsilon^L S_R - \epsilon^R S_L}{2(S_R - S_L)} \end{aligned} \quad (23)$$

Some interesting observations are forthcoming at this point. By comparing Eq. (23) with Eq. (21), we notice that both the modified coefficients $\bar{a}_0$ and $\bar{a}_1$ turns out respectively to be combinations of the unmodified coefficients a_0 and a_1 followed by additional terms that are functions of the dissipation parameter $\epsilon^{L,R}$. If we choose $\epsilon^R > \epsilon^L$ then, these terms are positive additions and we can ensure that $\bar{a}_0 > a_0$ and $\bar{a}_1 > a_1$. Note that such a constraint on $\epsilon^{L,R}$ can be imposed by simply setting say, $\epsilon^L \sim |S_L|$ and $\epsilon^R \sim |S_R|$ in a subsonic flow. We now consider a simple choice of $\epsilon^L = \epsilon^R = \epsilon$. Then, the additional term in $\bar{a}_0$ vanish while, the additional term in $\bar{a}_1$ becomes $\epsilon/2$. Thus, for such a choice, it is seen that the modification presented in Eq. (18) effectively affects only the $\mathbf{U}_L - \mathbf{U}_R$ component (associated with the coefficient a_1) of the dissipation term $\mathbf{D}_{HLL}$ and nowhere else in the entire HLLC formulation. This fact can be exploited to reduce the cost associated with implementation of the present cure to an already existing HLLC Riemann solver code.

The above discussion clearly indicates that increasing the magnitude of the HLL-type numerical dissipation term $\mathbf{D}_{HLL}$ present in the HLLC scheme, by enhancing the nonlinear wavespeed estimates S_L and S_R appearing in it, using the dissipation parameter $\epsilon^{L,R}$, provides an excellent framework to introduce supplementary numerical dissipation into the original

HLLC scheme. Referring to Eqs. (19) and (20), we realize that this additional numerical dissipation is made available only on the first and third Riemann flux components in our proposed HLLC-SWM framework. However, it remains to be seen whether this supplementary dissipation is sufficient to avoid shock instability plaguing the scheme. In the next section, we perform a linear perturbation analysis on the HLLC-SWM framework to investigate this concern.

6. Linear perturbation analysis of HLLC-SWM framework

The technique of predicting whether a numerical scheme will produce shock unstable solution based on how the scheme evolves an initially perturbed primitive variable profile, was established by Quirk [10] and improved by others [9,11]. In this section, we use this technique to study the effect of the dissipation parameter $\epsilon^{L,R}$ and its associated coefficient α on this evolution.

In the present work, we setup the initial perturbed flow field as follows. A steady mean flow is chosen with normalized state values of $\rho_0 = 1$, $u_0 \neq 0$, $v_0 = 0$ and $p_0 = 1$ for density, x-directional velocity, y-directional velocity and pressure respectively. Then, we choose a set of three adjoining cells lying in the transverse direction that exists in this mean flow and label them as ‘even’, ‘odd’ and ‘even’ consecutively. For a cell ‘j’ amongst them, the perturbed initial flow field is introduced as a saw-toothed profile by setting,

$$\begin{cases} \rho_j = \rho_0 + \hat{\rho}, & p_j = p_0 + \hat{p}, & u_j = u_0 + \hat{u} & \text{if } j \text{ is even} \\ \rho_j = \rho_0 - \hat{\rho}, & p_j = p_0 - \hat{p}, & u_j = u_0 - \hat{u} & \text{if } j \text{ is odd} \end{cases} \quad (24)$$

where, $\hat{\rho}$, $\hat{u}$ and $\hat{p}$ denote perturbations in density, x-directional velocity and pressure respectively. Note that we avoid introducing perturbations $\hat{v}$ in y-directional velocity in the present analysis because it has been reported that effect of this component is to dampen out perturbations in other components, eventually stabilizing the scheme [11,18]. Had it been included, then isolating the effect of parameter $\epsilon^{L,R}$ on other perturbations would have been difficult. For the HLLC scheme, the evolution equation for $\hat{\rho}$, $\hat{u}$ and $\hat{p}$ can be derived (following the procedure discussed in [11]) as,

$$\begin{aligned} \hat{\rho}^{n+1} &= \underbrace{\hat{\rho}^n(1-2\lambda)}_{\text{HLL component}} + \underbrace{2\lambda \left(\hat{\rho}^n - \frac{\hat{p}^n}{\gamma} \right)}_{\text{Antidiffusive component}} \\ \hat{u}^{n+1} &= \underbrace{\hat{u}^n(1-2\lambda)}_{\text{HLL component}} + \underbrace{2\lambda(\hat{u}^n)}_{\text{Antidiffusive component}} \\ \hat{p}^{n+1} &= \underbrace{\hat{p}^n(1-2\lambda)}_{\text{HLL component}} \end{aligned} \quad (25)$$

Corresponding evolution equations for the HLLC-SWM framework can be derived as,

$$\begin{aligned} \hat{\rho}^{n+1} &= \underbrace{\hat{\rho}^n(1-2\lambda)}_{\text{HLL component}} + \underbrace{2\lambda \left(\left(1 - \frac{\alpha \epsilon^{L/R}}{\sqrt{\gamma}} \right) \hat{\rho}^n - \frac{\hat{p}^n}{\gamma} \right)}_{\text{Antidiffusive component}} \\ \hat{u}^{n+1} &= \underbrace{\hat{u}^n(1-2\lambda)}_{\text{HLL component}} + \underbrace{2\lambda \left(1 - \frac{\alpha \epsilon^{L/R}}{\sqrt{\gamma}} \right) \hat{u}^n}_{\text{Antidiffusive component}} \\ \hat{p}^{n+1} &= \underbrace{\hat{p}^n(1-2\lambda)}_{\text{HLL component}} - \underbrace{2\lambda \left(\frac{\alpha \epsilon^{L/R}}{\sqrt{\gamma}} \right) \hat{p}^n}_{\text{Antidiffusive component}} \end{aligned} \quad (26)$$

where, $\lambda = \sqrt{\gamma} \frac{\Delta t}{\Delta x}$ denotes a linearized CFL number.

Notice from Eq. (25) for the HLLC scheme that, there exists independent contributions from the inherent HLL-type diffusive component and the antidiffusive component, towards the evolution of perturbations $\hat{\rho}$ and $\hat{u}$ while, contribution from the antidiffusive term is absent for the evolution of perturbation $\hat{p}$. The corresponding amplification factors for these perturbations are respectively $(1, 1, 1-2\lambda)$. A Von-Neumann type stability bound on λ can be obtained from these as,

$$0 < \lambda < 1 \quad (27)$$

Hence, for $0 < \lambda < 1$, the perturbations $\hat{\rho}$ and $\hat{u}$ are not damped by the HLLC scheme while the perturbation $\hat{p}$ is attenuated. However, although $\hat{p}$ gets attenuated with time, any finite $\hat{p}$ is repeatedly fed into $\hat{\rho}$, which results in existence of a residual

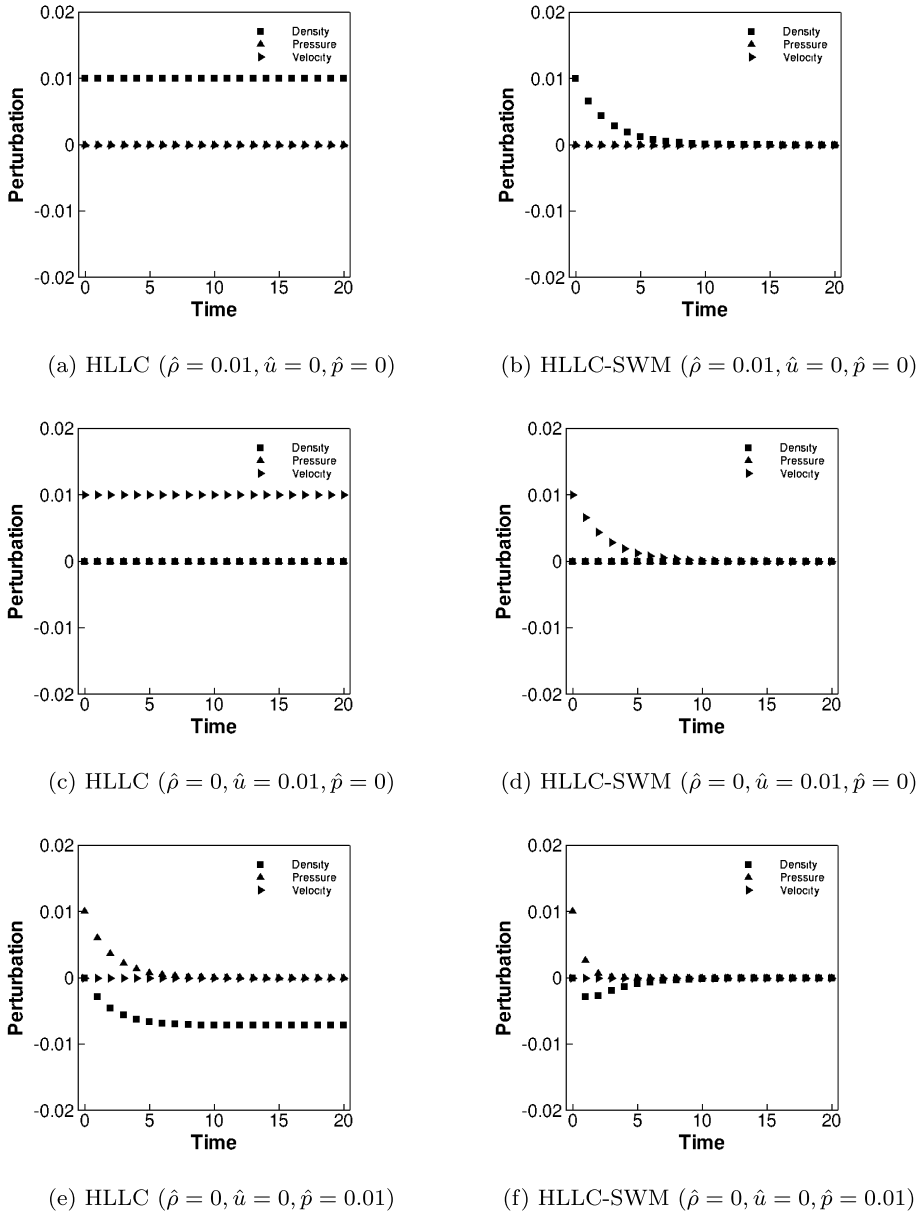


Fig. 1. Comparison of the evolution of density, x-velocity and pressure perturbations achieved by the HLLC scheme and the HLLC-SWM framework.

$\hat{\rho}$ that is growing in time. These behaviors can be observed in Figs. 1(a), 1(c) and 1(e) that form the left panel of Fig. 1. Each of these figures indicate the evolution of perturbation in either density, x-velocity or pressure for a given initial perturbation in one of those quantities (while setting the other initial perturbations to be 0). These results correspond to a value of $\lambda = 0.2$. The undamped perturbations in ρ and u variables will eventually lead to unphysical variation of the conserved quantity ρu in the vicinity of the shock and result in shock instability as discussed in [37]. Gressier et al. [9] have termed a scheme with at least one non-decaying amplification factor as *marginally stable scheme*. By due consideration, the HLLC scheme is a *marginally stable scheme*.

Now, examine Eq. (26) corresponding to the HLLC-SWM framework. Once again, we recognize independent contributions from the inherent HLL-type diffusive component and the antidiffusive component, towards the evolution of perturbations $\hat{\rho}$ and $\hat{u}$, along with an additional contribution to the evolution of perturbation $\hat{p}$. However, the terms arising from the antidiffusive component now contains the dissipation parameter $\epsilon^{L,R}$ and its associated coefficient α . The amplification factors for the perturbations $\hat{\rho}, \hat{u}, \hat{p}$ are respectively $(1 - 2\lambda \frac{\alpha \epsilon^{L,R}}{\sqrt{\gamma}}, 1 - 2\lambda \frac{\alpha \epsilon^{L,R}}{\sqrt{\gamma}}, 1 - 2\lambda(1 + \frac{\alpha \epsilon^{L,R}}{\sqrt{\gamma}}))$. These lead to the following stability bound on the linearized CFL number λ (also taking into account the usual stability criterion),

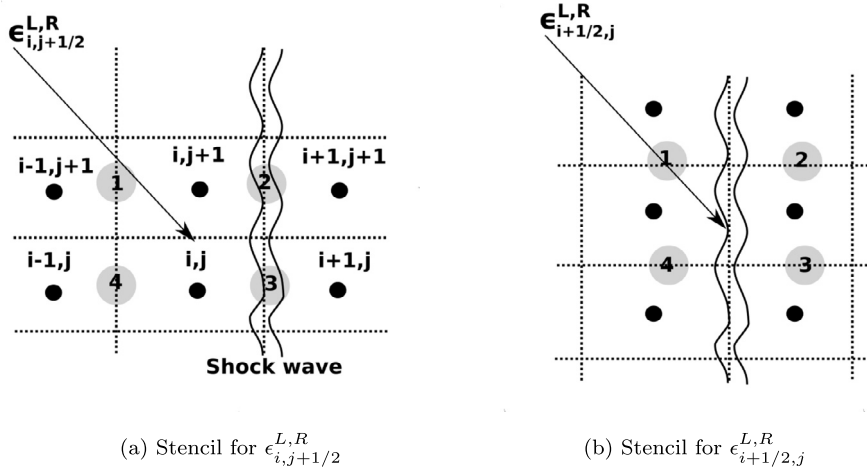


Fig. 2. Typical stencils used for evaluation of dissipation parameter (a) $\epsilon_{i,j+1/2}^{L,R}$ on transverse interfaces (b) $\epsilon_{i+1/2,j}^{L,R}$ on longitudinal interfaces.

$$0 < \lambda < \min\left(1, \frac{1}{1 + \frac{\alpha \epsilon^{L,R}}{\sqrt{\gamma}}}\right) \quad (28)$$

It can be verified that irrespective of the values of dissipation parameter $\epsilon^{L,R}$ and coefficient α , the quantity $\frac{1}{1 + \frac{\alpha \epsilon^{L,R}}{\sqrt{\gamma}}} < 1$.

Hence, the actual stability criterion which should be used while numerical computation will be governed by the factor $\frac{1}{1 + \frac{\alpha \epsilon^{L,R}}{\sqrt{\gamma}}}$. For any value for $\epsilon^{L,R}, \alpha > 0$ such that λ falls within in these bounds, it can be seen that an additional damping

factor of $2\lambda \frac{\alpha \epsilon^{L,R}}{\sqrt{\gamma}}$ is introduced into $\hat{\rho}$ and $\hat{u}$ perturbations. In case of $\hat{\rho}$, the additional damping factor ensures that any residual perturbations arising in $\hat{\rho}$, due to action of $\hat{p}$, are suppressed. The perturbations in $\hat{u}$ are also damped with time by the additional damping factor. Thus, in general, the HLLC-SWM framework ensures attenuation of the perturbations in all the three primitive quantities. These behaviors are evident in the right panel of Fig. 1. The plots correspond to $\lambda = 0.2$, $\alpha = 4$ and $\epsilon^{L,R} = 1.0$. Since the scheme has decaying amplification factors for all quantities, it can be regarded as a *strictly stable* scheme and is guaranteed to be shock instability free [9]. In the next section, we propose two strategies to estimate the dissipation parameter $\epsilon^{L,R}$.

7. Estimation of dissipation parameter $\epsilon^{L,R}$

The SWM framework provides an opportunity to construct a family of shock stable HLLC schemes depending on the choice of $\epsilon^{L,R}$. The purpose of $\epsilon^{L,R}$ is primarily to increment the value of $\overline{S_L}$ and $\overline{S_R}$, appearing in the inherently-present HLL-type numerical dissipation term $\mathbf{D}_{HLL}$, in the vicinity of a shock wave and tend to zero elsewhere. Hence, any viable shock sensor is a good candidate for it. Since shock instability is primarily a multidimensional phenomenon, it is also advisable to have an estimation of $\epsilon^{L,R}$ that takes this fact into account [17]. Below we discuss some strategies to achieve a satisfactory estimate of $\epsilon^{L,R}$ based on these considerations.

7.1. Eigenvalues based estimation of $\epsilon^{L,R}$

One possible estimate for $\epsilon^{L,R}$ can be based upon the idea of an entropy-fix. Several authors [17,11,26,27] have used such entropy-fix-based estimates to save the Roe scheme from shock instability. Following [11], we define $\epsilon^{L,R}$ at an interface $(i, j + 1/2)$ as,

$$\epsilon_{i,j+1/2}^{L,R} = \max(\eta_1, \eta_2, \eta_3, \eta_4) \quad (29)$$

where, each η_k ($k = 1..4$) corresponds to the dissipation quantity calculated at predefined interfaces that form a stencil around the interface $(i, j + 1/2)$ as shown in Fig. 2(a). Each η_k can be in turn evaluated using a common entropy fix formula,

$$\eta_k = \frac{1}{2} \max(|\lambda_p(\mathbf{U}_R) - \lambda_p(\mathbf{U}_L)|) \quad (30)$$

where, p iterates over all the characteristic wave speeds λ_p of the Riemann problem at k th interface. It can be shown that this is equivalent to simply evaluating,

$$\eta_k = \frac{1}{2} \max(|u_{nR} - u_{nL}| + |a_R - a_L|) \quad (31)$$

which turns out to be the maximum of difference between the speeds of the rightmost running nonlinear wave of the right states and the leftmost running nonlinear wave of the left states across all the interfaces comprising the stencil. It is important to state here that the idea of an entropy fix is utilized in the strategy only to determine an estimate for $\epsilon^{L,R}$ that is $\mathcal{O}(S_L, S_R)$. Since the HLLC-SWM framework builds upon the HLLC scheme, which by default is entropy satisfying, it does not require such fixes to ensure entropy satisfying solutions. An $\epsilon^{L,R}$ at any interface defined in the above form represents the largest one-dimensional entropy correction corresponding to all the associated interfaces that forms a predefined stencil around it. It works both as an estimate for the quantity of dissipation to be introduced and also as a shock sensor (since it is designed to detect change in characteristic speeds, which will indicate a shock front). A corresponding definition for $\epsilon_{i+1/2,j}^{L,R}$ at $(i + 1/2, j)$ interface can be arrived at by choosing a stencil as shown in Fig. 2(b), while retaining the definition of η_k 's. The stencils for implementation are carefully chosen so that $\epsilon^{L,R}$ will be most active only on the critical transverse interfaces adjoining the shock front, while remaining negligible on other interfaces of the computational grid. Further, as pointed out in [11], the stencils also help in avoiding extraneous dissipation from being evaluated when computing flow features other than shocks, with naturally occurring gradients, like the boundary layer. This is primarily achieved by excluding the $(i, j + 1/2)$ th (or equivalently the $(i + 1/2, j)$ th) interface itself from the stencil used to calculate $\epsilon^{L,R}$. Further, the dissipation provided through $\epsilon^{L,R}$ has a multidimensional flavor because the dissipation introduced at a longitudinal interface is a function of the solution variation in the transverse direction and vice versa [17]. This allows infusion of requisite amount of cross-flow dissipation in the vicinity of the shock front which potentially guarantees shock stability. An HLLC-SWM framework that uses this estimate for $\epsilon^{L,R}$ is hereby termed as HLLC-SWM-E (Eigenvalue based) scheme.

7.2. Pressure based estimation of $\epsilon^{L,R}$

Since the entropy-fix-based estimation of $\epsilon^{L,R}$ works both as a shock sensor and as an estimate of the quantity of dissipation to be provided at it, it can lead to slight amount of dissipation in flows with strong velocity gradients. An interesting way to overcome this would be to introduce a separate function for shock sensing that is capable of distinguishing between a shock discontinuity and other types of waves. Pressure-ratio-based functions are appropriate in this regard since, the pressure variable undergoes changes only across shocks while remaining unchanged across contact and shear waves. Hence, we redefine $\epsilon^{L,R}$ as,

$$\epsilon_{i,j+1/2}^{L,R} = (1 - \omega_{i,j+1/2}) \max(\eta_1, \eta_2, \eta_3, \eta_4) \quad (32)$$

where, $\omega_{i,j+1/2}$ is a pressure-ratio-based shock sensor that can be constructed as [38],

$$\omega_{i,j+1/2} = \min(f_1, f_2, f_3, f_4) \quad (33)$$

where, f_k 's are pressure-ratio-based functions evaluated on a predefined stencil around $(i, j + 1/2)$ interface similar to the one shown in Fig. 2(a). At k th interface, f_k is defined as,

$$f_k = \min \left(\frac{p_R}{p_L}, \frac{p_L}{p_R} \right)_k^\beta \quad (34)$$

Here, p_R and p_L denotes the right and left cell-center pressures across k th interface. A symmetrical definition holds for $\epsilon_{i+1/2,j}^{L,R}$. A value of $\beta = 5.0$ is used. Note that this pressure-ratio-based $\epsilon^{L,R}$ also retains the merits of the shock sensor discussed in Sec. 7.1 i.e., a multidimensional stencil and local solution dependence. An HLLC-SWM scheme that uses this kind of $\epsilon^{L,R}$ will hereby be addressed as HLLC-SWM-P (Pressure based) scheme. In the next section, we complete the description of the HLLC-SWM framework by estimating an appropriate value for the parameter α .

8. Estimation of dissipation parameter α

The dissipation parameter α controls the extent to which the wave speeds S_L and S_R are modified by the $\epsilon^{L,R}$ parameter. Too large a value of α could result in larger magnitudes of $\bar{S}_L$ and $\bar{S}_R$ which, as seen from Eq. (23), will result in larger magnitude of $\bar{\mathbf{D}}_{HLL}$ and eventually threaten to spoil the accuracy of the solution. On the other hand, too small a value could fail to provide adequate numerical dissipation to counter the threat of shock instability. Thus, choice of α plays a crucial role in the overall effectiveness of the HLLC-SWM class of schemes and hence must be carefully chosen. We envision that a reasonable lower bound on α , that is capable of dealing with shock instability, can be obtained by studying the effect of varying α on the stability of a shock profile corresponding to a simple shock instability test problem: the steady simulation of a two-dimensional isolated standing shock. Ideally, the steady state solution of this problem will consist of the original unperturbed shock profile that will be preserved forever. However, the presence of shock instability would immediately manifest as unsteady perturbations in the physical quantities in the aftershock region. Since, the shock strength (represented by free stream inlet Mach number) is a crucial parameter affecting shock instability behavior of a scheme [7], we would like to obtain a choice of α that is capable of dealing with the instability over a wide range of Mach numbers.

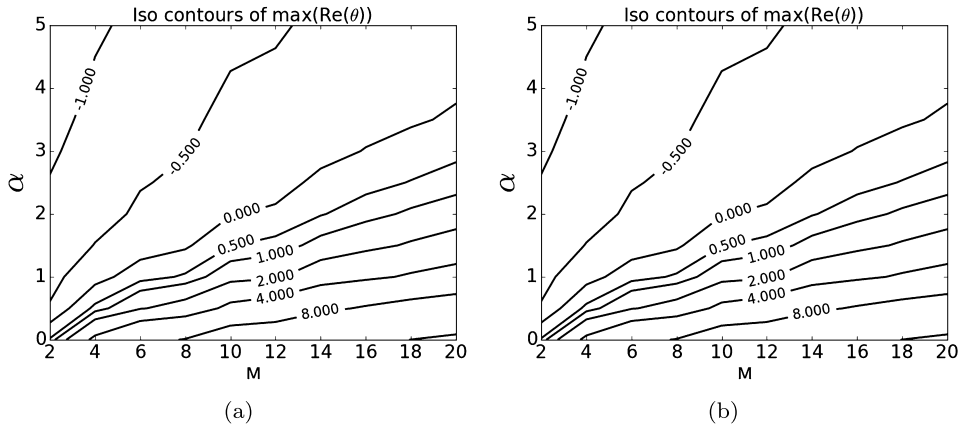


Fig. 3. Figure showing iso-contours of $\max(\text{Re}(\theta))$ on an $M - \alpha$ plot for the problem of a two-dimensional isolated standing shock computed using (a) the HLLC-SWM-E and (b) the HLLC-SWM-P schemes.

To this end, in the following study, we use a range of Mach numbers from $M = 2$ to $M = 20$. However, verifying the instability behavior through actual numerical simulation, for all the cases, would be cumbersome and hence, we resort to the technique of matrix based stability analysis as proposed by Dumsber et al. [7]. In the present work, the initial shock is located between the 5th and 6th column of cells in a regular Cartesian grid comprising of 11×11 cells on a 1.0×1.0 domain. The initial conditions on the supersonic side of the shock are chosen as $(\rho, u, v, p)_L = (1.0, 1.0, 0.0, 0.01457)$ while the Rankine–Hugoniot conditions are used to determine the corresponding conditions on the subsonic side. Periodic conditions are applied on the top and bottom boundaries. Since the HLLC scheme exactly preserves a thin shock, the initial and boundary conditions can be exactly specified [7]. The eigenvalue problem is solved using the eigenvalue function $\text{eig}()$ from *linalg* package in Python version 2.7.6.

In Fig. 3, we show the iso-contours of the max Real part of eigenvalues ' θ ' (denoted henceforth as $\max(\text{Re}(\theta))$), computed using both the HLLC-SWM-E and HLLC-SWM-P schemes, presented on an $M - \alpha$ plot. Since $\max(\text{Re}(\theta))$ represents the largest error growth rate, all $\max(\text{Re}(\theta))$ values shown in Fig. 3 belonging to the region lying below the contour line denoting '0.0' corresponds to unstable (exponentially amplifying) solutions while, all $\max(\text{Re}(\theta))$ values belonging to the region lying above it represents shock stable (exponentially decaying) solutions on this problem. Further, an $\alpha = 0.0$, which coincides with the x -axis, corresponds to the unmodified HLLC scheme which is known to be unstable on this problem for the range of Mach numbers tested.

Firstly, notice that the both variants of the HLLC-SWM framework have similar behavior on this experiment. This is expected since, these variants differ only by the pressure-ratio-based shock sensor term $(1 - \omega_{i,j+1/2})$, which maintains a value close to unity in the vicinity of a shock wave, allowing both the schemes to acquire similar magnitude for their respective dissipation parameter $\epsilon^{L,R}$. It is observed that for both the schemes, at any given Mach number, increasing α results in a reduction in $\max(\text{Re}(\theta))$ which indicates a tendency to establish a shock stable solution. For small Mach numbers, α value required to ensure linear stability is smaller than that required by a larger Mach number. This is an expected trend since, it is already known that stronger shocks lead to faster manifestation of instability [7]. We draw attention to the fact that an $\alpha = 4.0$ seems to guarantee linear shock stability for the entire range of Mach numbers tested and will be used to configure both variants of the HLLC-SWM schemes.

We attempt to verify these predictions by resorting to a numerical simulation of the above problem for a specific case of $M = 10$ computed using the HLLC-SWM-E scheme. Fig. 4 shows variation of the cell-centered density values in the y -direction extracted right behind the shock, i.e. from the 7th column of cells in the computational domain for $\alpha = 0.0, 1.0, 2.0$ and 4.0 . Since the problem has an exact steady state solution, any deviation in the value of the density variable from its exact steady value of 5.44 clearly indicates presence of instability. The solutions are obtained at $t = 20$ units. Fig. 4(a), corresponding to $\alpha = 0.0$, clearly shows the saw-tooth-profiled density variation that is typical of a shock unstable solution [10]. Further, as predicted by the analysis, increasing α values are seen to stabilize the solution. For $\alpha \geq 2.0$ the HLLC-SWM-E scheme is able to infuse sufficient dissipation to render a shock stable solution on this problem which agrees well with the prediction.

Lastly, we present the eigenvalue spectrums corresponding to these schemes for a representative Mach number of $M = 7$ in Fig. 5. The corresponding spectrum for the original HLLC is also provided for comparison. The eigenvalue spectrum visibly shows how the proposed HLLC-SWM-E and the HLLC-SWM-P schemes utilize the modified dissipation $\bar{\mathbf{D}}_{HLL}$ to shift the unstable error growth rates to the stable region. While the HLLC scheme has the largest error growth rate of +8.15562, the HLLC-SWM-E and the HLLC-SWM-P schemes, with $\alpha = 4.0$, are able to bring this down to -0.66053 and -0.59317 respectively. Amongst the HLLC-SWM schemes, the HLLC-SWM-E scheme turns out to be more stable.

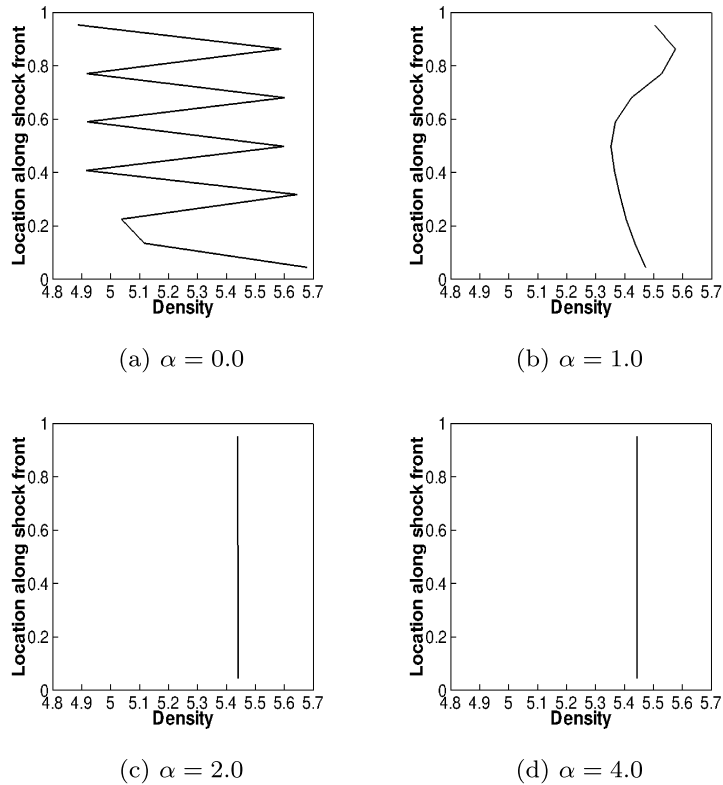


Fig. 4. Variation of density along the computed shock front extracted from the 7th column of cells in case of a $M = 10$ two-dimensional standing shock problem. The solution was obtained using the proposed HLLC-SWM-E scheme for selected α values. The deviation in the value of density variable from its exact steady value of 5.44 indicates presence of instability. The solutions are obtained at $t = 20$ units.

9. Numerical verification of the HLLC-SWM schemes

In this section, we demonstrate the efficacy of the proposed HLLC-SWM schemes through a series of strict test problems on which the original HLLC scheme is known to produce shock unstable results. It has been reported in [9] that the intensity of instability manifestation in certain test problems reduces as the order of accuracy of the computation increases, although it does not completely remove it. Based on this, certain test cases are computed to first order accuracy while, second order solutions are sought for others to clearly showcase the improvement in the solution obtained using the proposed schemes. A second order spatial accuracy in primitive variables is achieved by limiting gradients obtained using the Green Gauss method [39] with the Barth Jersperson limiter [40]. A second order time accuracy is achieved using strong stability preserving variant of the Runge Kutta method [41]. All boundary conditions are set using ghost cells. A value of $\alpha = 4.0$ is used to configure both HLLC-SWM schemes.

9.1. Moving shock instability problem

Shock instability that occurs in a $M = 6$ moving shock, propagating down a computational tube filled with a stationary fluid defined by $(\rho, u, v, p) = (1.4, 1.0, 0.0, 1.0)$, was first reported by Quirk [10]. The instability is triggered by perturbing the centerline grid of the computational domain to an order of $1E-6$. First order solution is sought. The CFL number for the computations were taken to be 0.5 and simulations were run till $t = 120$. The results showing fifty density contours equally spanning values from 1.4 to 7.34 are shown in Fig. 6. Fig. 6(a) clearly shows the deteriorated condition of the shock profile calculated by the original HLLC scheme. Figs. 6(b), 6(c) demonstrate that both the HLLC-SWM-E and the HLLC-SWM-P schemes are quite capable of suppressing the instability on this problem and maintaining a clean shock profile.

9.2. Stationary inclined shock instability problem

This problem was reported in [42]. A shock wave of strength $M = 7$ is setup along the line $y = 2(x - 12)$ at an angle of 63.43° with respect to the global x -direction. The computation mesh consists of 50×30 cells uniformly spanning a domain of size 50.0×30.0 . The left and right boundaries are maintained as supersonic inlet and subsonic outlet respectively while, periodic conditions are applied to the top and bottom walls. The CFL number for the computations were chosen as 0.5 and

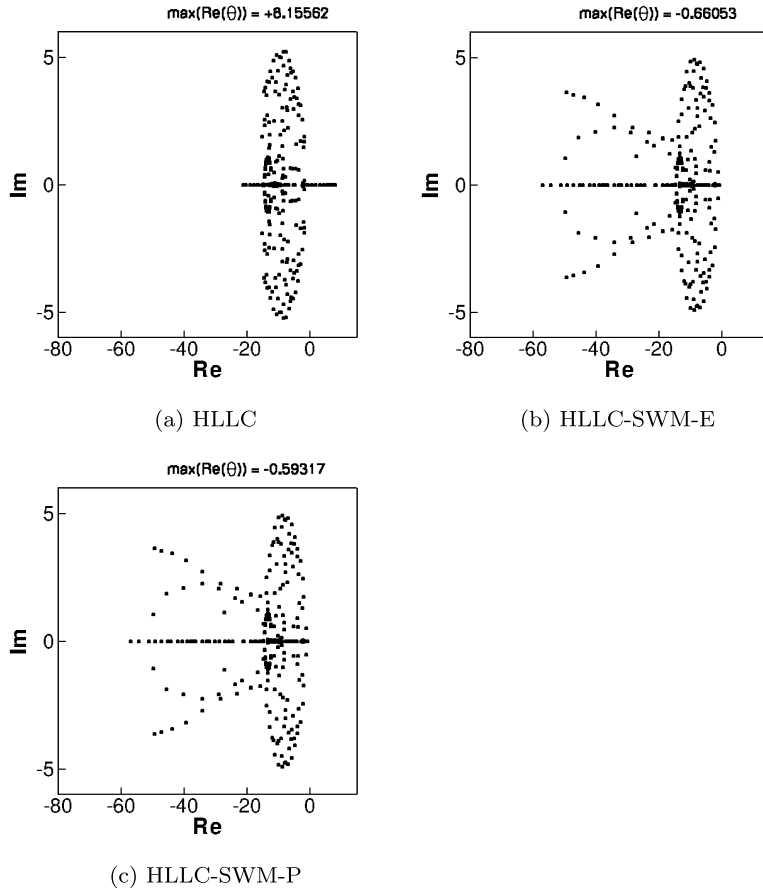


Fig. 5. Eigenvalue spectrums for the $M = 7$ two-dimensional standing shock problem obtained from matrix analysis (a) HLLC scheme (b) HLLC-SWM-E scheme (c) HLLC-SWM-P scheme. The parameter α is taken to be 4.0 for both the HLLC-SWM-E and the HLLC-SWM-P schemes.

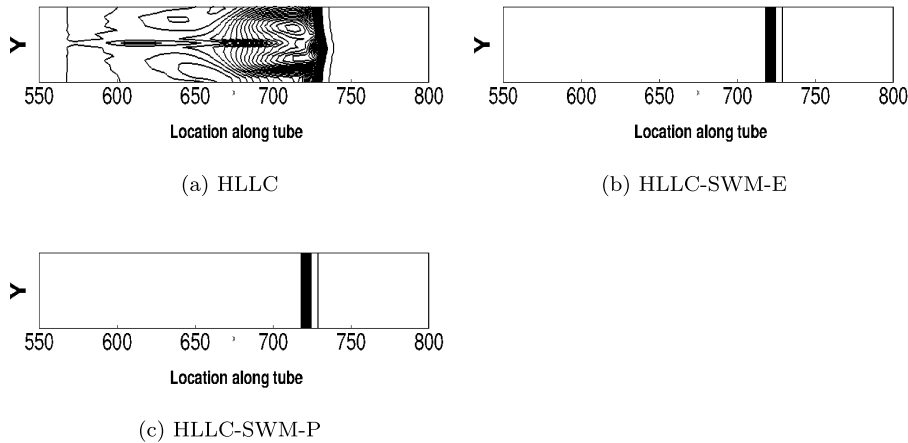


Fig. 6. Results showing fifty density contours equally spanning values from 1.4 to 7.34 for the $M = 6$ moving shock instability problem computed by the HLLC, the HLLC-SWM-E and the HLLC-SWM-P schemes. Results shown at $t = 120$.

first order solution was sought. Fig. 7 shows the results of this experiment at $t = 5$ where thirty density contours uniformly spanning values from 1.0 to 5.44 are shown.

Fig. 7(a) shows the presence of instability in the HLLC scheme. Fig. 8(a) shows the density variation along the shock front extracted just behind the shock. The density variation has a typical sawtooth profile. Fig. 7(b) shows the density contours computed by the HLLC-SWM-E scheme. It was noticed that although till $t = 2$ units there was no trace of instability in the computed flow field, some form of perturbations were observed in the simulation later. These perturbations however

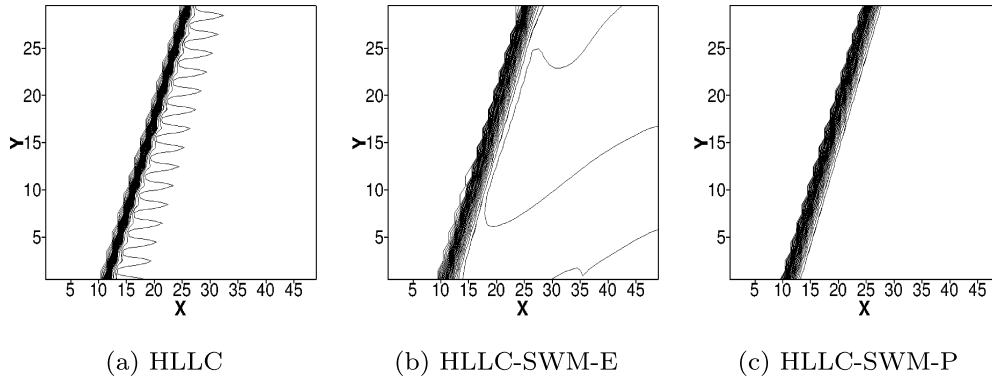


Fig. 7. Results showing thirty density contours equally spanning values from 1.4 to 5.44 for the $M = 7$ stationary inclined shock instability problem computed by the HLLC, the HLLC-SWM-E and the HLLC-SWM-P schemes. Results shown at $t = 5$.

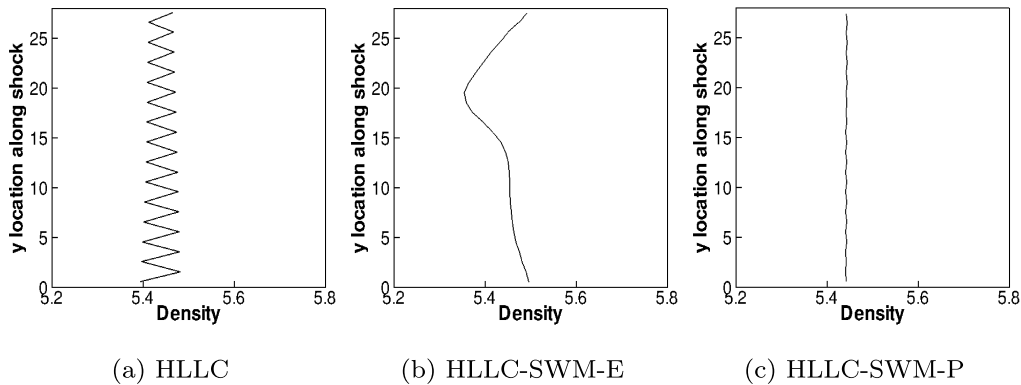


Fig. 8. Figure showing variation of density values in the immediate subsonic region along the shock profile for the $M = 7$ stationary inclined shock instability problem computed by the HLLC, the HLLC-SWM-E and the HLLC-SWM-P schemes. Results shown at $t = 5$.

remained temporally bounded and did not degrade the shock profile. The respective density variation is shown in Fig. 8(b). It can be seen that the nature of these perturbations are not similar to the spatially oscillatory one observed for the HLLC scheme and hence, may not indicate instability. Fig. 7(c) shows the density contours computed by the HLLC-SWM-P scheme. The corresponding density variation is shown in Fig. 8(c). It can be seen that introduction of the pressure sensor helps the HLLC-SWM-P scheme to outperform the HLLC-SWM-E scheme on this test case. The instability has been drastically minimized and the computed field is preserved very close to the exact solution.

9.3. Supersonic flow over a bluntbody

The steady state numerical solution of a supersonic flow over cylindrical body is a routine test problem to investigate the susceptibility of numerical schemes to a particularly dreaded form of shock instability: the Carbuncle phenomenon. This failure is characterized by visible distortions that appear in the expected smooth bow shock profile in front of the body. In this work, a cylindrical body of unit radius is subjected to a $M = 20$ flow. We use 320×20 body fitted structured quadrilateral cells in circumferential and radial directions respectively. Detailed problem description can be found in [22]. The computation of this problem is carried out to first order accuracy. The CFL number for the computations were taken to be 0.5 and the simulations were run for 80,000 iterations. The results showing twenty density contours equally spanning value from 1.4 to 8.5 is shown in Fig. 9.

Fig. 9(a) clearly shows the onset of the instability in the shock profile computed by the HLLC scheme. Figs. 9(b) and 9(c), on the other hand, demonstrates that both variants of the HLLC-SWM framework are quite capable of computing clean shock profiles, and post-shock subsonic regions that are free of the instability.

We probe the quality of these solutions further by extracting the centerline pressure data along the stagnation line $j = 160$ as shown in Fig. 10(a). We aim to verify the accuracy of these predictions by comparing them with the exact inviscid value for this problem at two specific locations: at a point lying immediately behind the normal shock region of the captured bow shock and at the stagnation point. These selected locations are visible in regions 'A' and 'B' marked in Fig. 10(a) and are shown magnified in Figs. 10(b) and 10(c). In general, we note that all the three schemes capture shocks that are spread over two columns of cells. Referring to Fig. 10(b), we notice that the HLLC scheme predicts a pressure

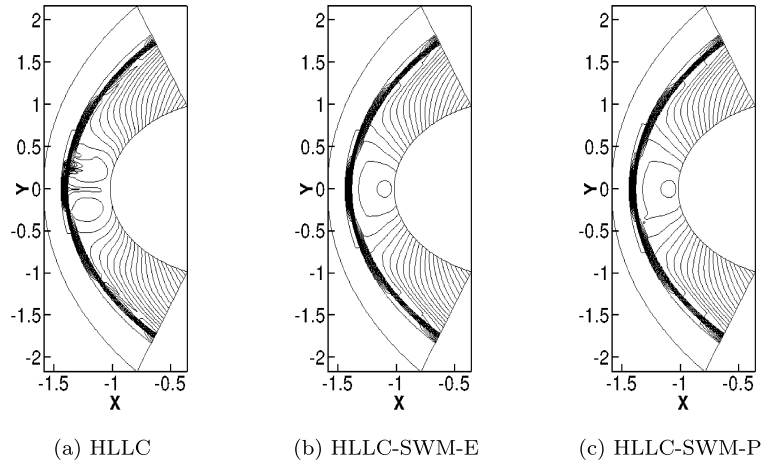


Fig. 9. The results showing twenty density contours equally spanning value from 1.4 to 8.5 for the $M = 20$ supersonic flow over a blunt body computed by the HLLC, the HLLC-SWM-E and the HLLC-SWM-P schemes. Results are shown after 80000 iterations.

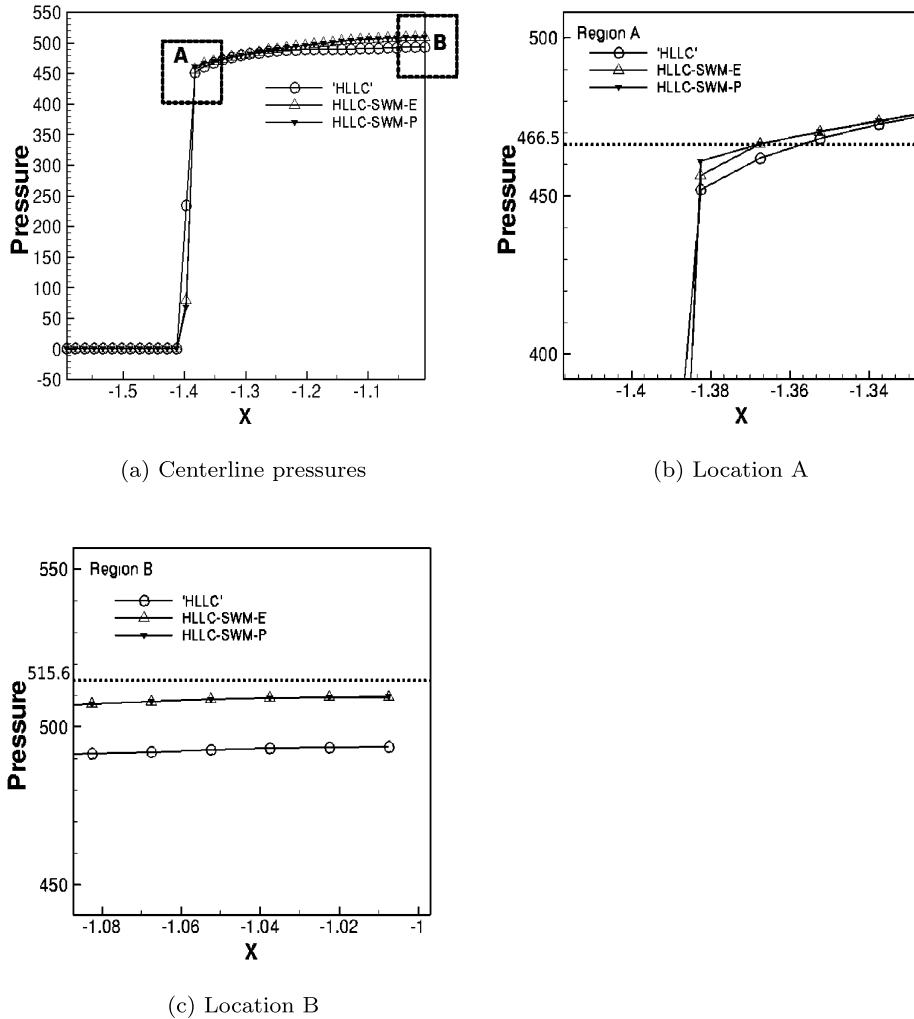


Fig. 10. Comparison of centerline pressure values for the $M = 20$ supersonic flow over a blunt body computed by the HLLC, the HLLC-SWM-E and the HLLC-SWM-P schemes. Results are shown after 80000 iterations.

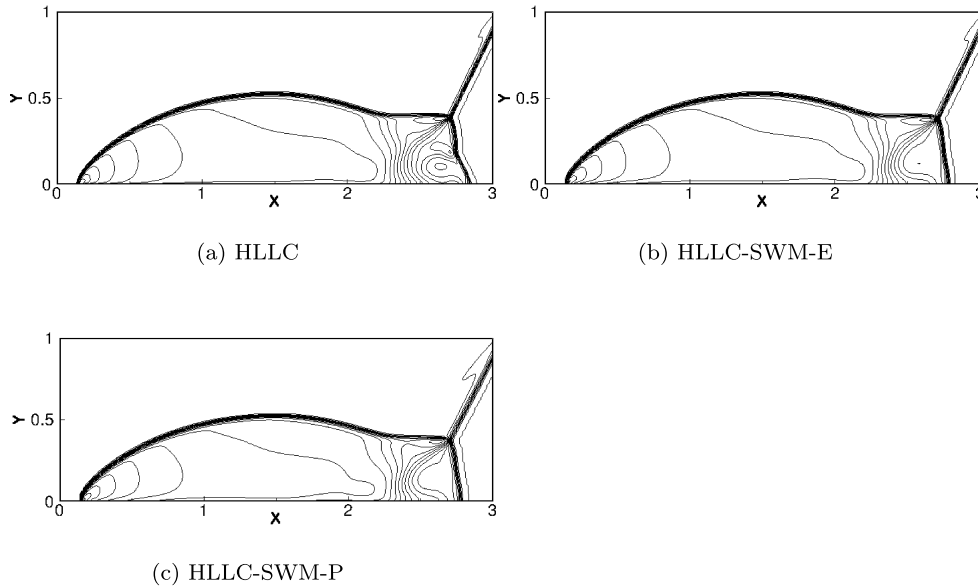


Fig. 11. Results showing twenty-five equispaced density contours spanning values from 1.4 to 21 for the $M = 10$ Double Mach Reflection problem computed by the HLLC, the HLLC-SWM-E and the HLLC-SWM-P schemes. Results are shown at $t = 0.2$.

value of 461.83 Pa at a point which lies immediately behind the normal shock region of the bow shock, which is lesser than the exact value of 466.5 Pa at this location. On the other hand, the HLLC-SWM-E and HLLC-SWM-P schemes predicts values of 466.28 Pa and 466.40 Pa respectively at this point which are closer to the exact value. Referring to Fig. 10(c), we notice that the HLLC scheme predicts a stagnation pressure of 493.72 Pa which is grossly underpredicted in comparison to the exact value of 515.6 Pa. The stagnation pressure predicted by the HLLC-SWM-E and HLLC-SWM-P schemes are respectively 509.50 Pa and 509.53 Pa which are much closer to the exact value. This example shows the fidelity of the proposed HLLC-SWM schemes in computing accurate, Carbuncle free solutions on such severe grids.

9.4. Double Mach Reflection problem

To demonstrate the robustness of the proposed schemes, another standard test problem called the Double Mach Reflection problem is used [10,43]. An oblique shock corresponding to $M = 10$, making a 60° angle with the bottom wall, is made to propagate through a domain of size 4.0×1.0 divided into 480×120 structured Cartesian cells containing a stationary fluid. Detailed problem description can be found in [43]. First order solution is sought for this case. The simulation were run till $t = 0.2$ with a CFL number of 0.7. Fig. 11 show results of the experiment with twenty-five equispaced density contours spanning values from 1.4 to 21.

In Fig. 11(a), the presence of kinked Mach stem produced by the HLLC scheme is clearly visible. However, both variants of the HLLC-SWM framework produce clean shock profiles which are devoid of these unphysical kinks.

9.5. Diffraction of a moving normal shock over a 90° corner

The problem of a Mach 5.09 normal shock's sudden expansion around a 90° corner was studied in [10]. We use a unit-dimensional domain meshed with 400×400 cells. The right angled corner, where the initial normal shock stands, is located at $x = 0.05$, $y = 0.45$. The problem is computed to second order accuracy. Simulations were run till $t = 0.1561$ units with a CFL number of 0.4. Thirty density contours equally spanning values from 0 to 7 are shown in Fig. 12.

Fig. 12(a) clearly demonstrates the extent of instability present in the computed shock profile produced by the HLLC scheme. A part of the normal shock in the right corner is completely distorted in this case. However, both variants of the HLLC-SWM framework are able to satisfactorily compute the solution without producing any trace of instability in the normal shock.

9.6. Supersonic flow over a forward facing step

A $M = 3$ supersonic flow over forward facing step is another standard problem that has been extensively studied earlier in [43]. The computational domain is of size 3×1 and is meshed with 120×40 structured Cartesian cells. A 0.2 units high step is located at a distance of 0.6 units from the inlet. The problem is computed to second order accuracy. The simulations were run till $t = 4$ with a CFL number of 0.4. Fig. 13 shows forty equispaced density contours spanning values from 0.2 to 7.0.

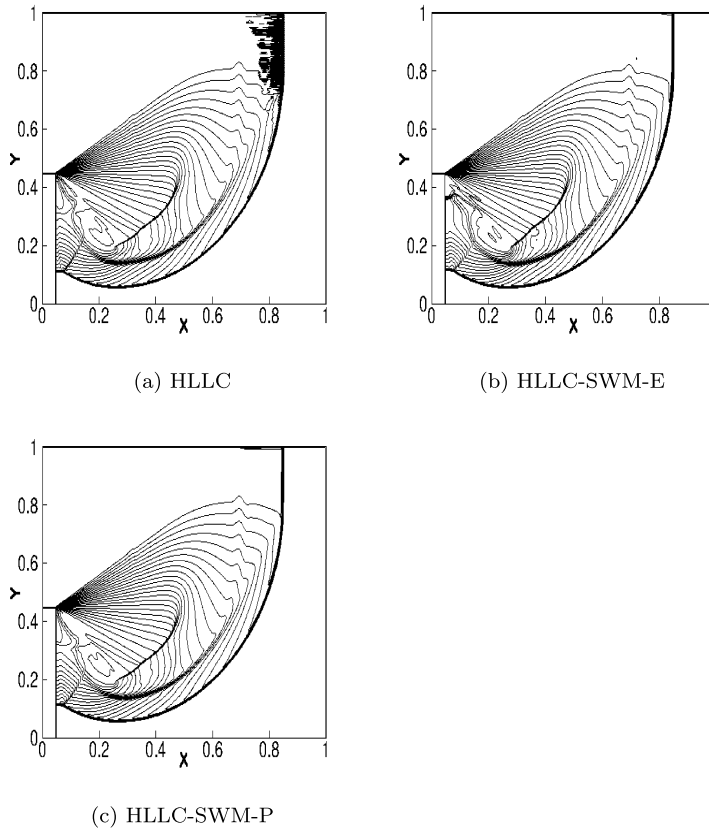


Fig. 12. Results showing thirty density contours equally spanning values from 0 to 7 for the $M = 5.09$ normal shock diffraction around a 90° corner computed by the HLLC, the HLLC-SWM-E and the HLLC-SWM-P schemes. Results are shown at $t = 0.1561$.

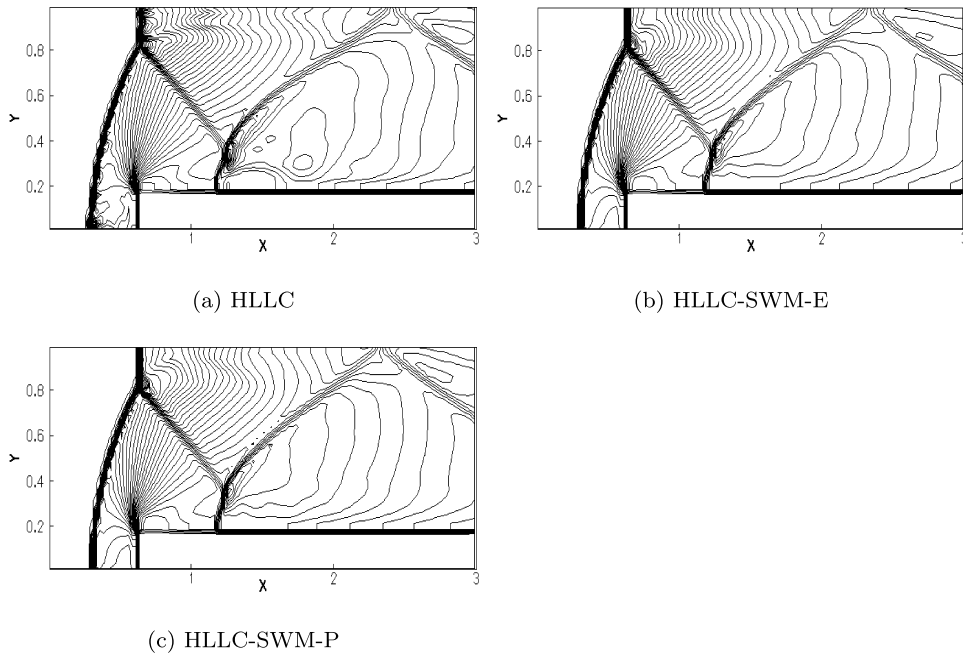


Fig. 13. Results showing forty equispaced density contours spanning values from 0.2 to 7.0 for the $M = 3$ supersonic flow over a forward facing step problem computed by the HLLC, the HLLC-SWM-E and the HLLC-SWM-P schemes. Results are shown at $t = 4$.

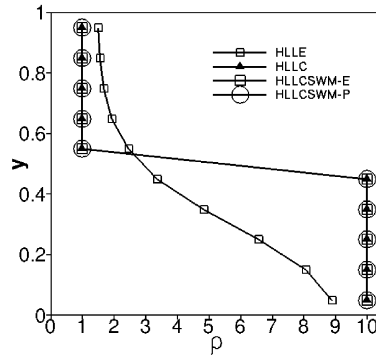


Fig. 14. Figure showing a comparison of density variation along the y -direction, at the center of the domain, for the two-dimensional supersonic shear flow computed by the HLLC, the HLLC-SWM-E and the HLLC-SWM-P schemes. Results are shown after 1000 iterations.

It is seen from Fig. 13(a) that predominant oscillations are visible in the shock profile computed by HLLC scheme. These are most prominent at the normal shock that exist as a part of the bow shock situated in front of the body and the normal shock stem appearing near the top wall. Further, HLLC solution also has a severely kinked reflected shock stem that may cause incorrect jumps in physical quantities across it on the body. In comparison, solution computed by both variants of the HLLC-SWM scheme, shown in Figs. 13(b), 13(c), depict clean shock profiles with no remnants of the instability visible. The infamous kink has also improved considerably in these solutions.

9.7. Two-dimensional inviscid shear flow

In this section we verify whether the proposed schemes retain the invaluable property of inviscid contact preservation which is a prerequisite for accurate resolution of viscous flows. The problem describes two fluids with different densities sliding at different speeds over each other and investigates the inviscid contact capturing ability of a given scheme [44]. The conditions for the top and bottom fluids are chosen as $(\rho, p, M)_{top} = (1.0, 1.0, 2.0)$ and $(\rho, p, M)_{bottom} = (10.0, 1.0, 1.1)$ respectively. A coarse grid consisting of 10×10 cells on a domain of 1.0×1.0 is used. The CFL number for the simulations are taken to be 1.0 and simulations are run for 1000 iterations. All simulations are computed to first order accuracy. To clearly distinguish between the behaviors of contact preserving and contact dissipative schemes, the solution for the shear dissipative HLLC scheme is also included for comparison. In Fig. 14 we show the density variation along y -direction at the center of the domain for comparison. From Fig. 14 it is clear all the schemes, except the HLLC scheme, are able to exactly capture the inviscid contact of this problem without any dissipation. Thus, it can be inferred that the modified HLL-type numerical dissipation term $\bar{\mathbf{D}}_{HLL}$, that introduces additional numerical dissipation into the HLLC scheme to deal with shock instability, has not affected the ability to resolve inviscid contact discontinuities.

9.8. Subsonic laminar flow over a flat plate

The original HLLC scheme is well known for its ability to accurately resolve shear dominated viscous flow features including the practically significant boundary layer flows. Hence, it is important to investigate how introduction of the modified dissipation term $\bar{\mathbf{D}}_{HLL}$ will affect the performance of the HLLC scheme on such flows. To perform this study, the test case of a $M = 0.1$ subsonic laminar flow of air at atmospheric conditions, over a flat plate of length $L = 0.3048$ m, is chosen. The total length of the domain is 0.381 m in x direction and 0.1 m in y direction. The domain is divided into 31×33 Cartesian cells with 15 cells within the boundary layer. Viscous fluxes were discretized using simple averaging. The CFL number was taken to be 0.7. The flow is considered to have achieved steady state when the horizontal velocity residuals dropped to the order of 10^{-7} . The normalized longitudinal velocity profiles ($\frac{u}{u_\infty}$) are plotted against the Blasius parameter $\eta = y\sqrt{u_\infty/\mu L}$ in Fig. 15. Further, in Fig. 16 we show the variation of the skin friction coefficient (C_f) along the length of the plate. In both these figures, we additionally show results computed by the highly dissipative HLLC scheme for comparison.

It is seen from these figures that among the proposed schemes, the HLLC-SWM-E scheme suffers a slight loss of accuracy within the boundary layer. The dissipative behavior of the HLLC-SWM-E scheme was expected since, the $\epsilon^{L,R}$ in Eq. (18) is based on computing the maximum difference in wave speeds at any interface. Hence, at any interface where gradients $\Delta_x(u, v)$ and $\Delta_y(u, v)$ are non-zero, $\epsilon^{L,R} \neq 0$. This would result in engaging of the modified numerical dissipation term $\bar{\mathbf{D}}_{HLL}$ on these interfaces, which would in turn introduce unwanted numerical dissipation and diffuse the boundary layer. On the other hand, the HLLC-SWM-P scheme, which uses a pressure-ratio-based shock sensor, is remarkably accurate everywhere within the boundary layer and exactly matches the Blasius profile.

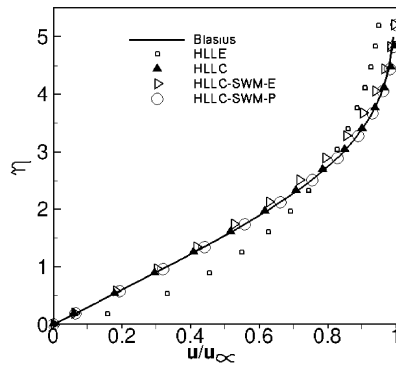


Fig. 15. Variation of the normalized longitudinal velocity ($\frac{u}{u_\infty}$) with Blasius parameter $\eta = y\sqrt{u_\infty/\mu L}$ for the $M = 0.1$ subsonic laminar flow over a flat plate computed by the HLLC, the HLLC-SWM-E scheme, the HLLC-SWM-P scheme and the HLLC scheme.

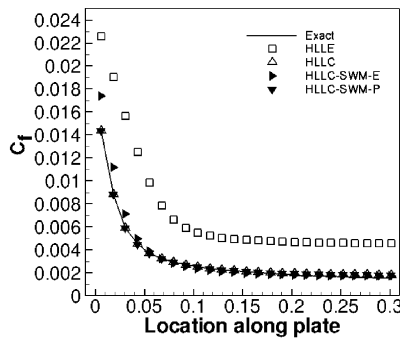


Fig. 16. Variation of the skin friction coefficient (C_f) along the length of the plate for the $M = 0.1$ subsonic laminar flow over a flat plate computed by the HLLC, the HLLC-SWM-E scheme, the HLLC-SWM-P scheme and the HLLC scheme.

10. Concluding remarks

In this paper, we proposed a simple cure for numerical shock instability phenomenon plaguing the Harten–Lax–van Leer with contact (HLLC) approximate Riemann solver. Firstly, we showed that the HLLC scheme can be split into a HLL-type diffusive component and an antidiffusive component responsible for restoring accuracy on the contact and shear waves. Once identified, we suggested a simple strategy to increase the magnitude of this HLL-type diffusive component, by enhancing the magnitudes of the nonlinear wavespeed estimates appearing in its numerical dissipation term $\mathbf{D}_{HLL}$, through a dissipation parameter ϵ and an associated coefficient α . The resulting enhanced numerical dissipation term $\bar{\mathbf{D}}_{HLL}$, need to be applied only on the first and third Riemann flux components for it to be effective in preventing shock instability. This strategy is termed as **Selective Wave Modification** and the HLLC scheme incorporating it was addressed as the HLLC-SWM scheme. We derived perturbation evolution models for this framework and showed that the SWM strategy essentially introduced additional damping terms which were a function of ϵ into the perturbation evolution equations for ρ and u fields. This guaranteed consistency of the conserved quantity ρu both along and across the shock and encouraged shock stability. However, this strategy was found to impose additional stability criterion on the CFL number associated with its use which was slightly more restrictive than the usual stability criterion.

To estimate the magnitude of ϵ , we proposed two entropy fix based approaches. Both of them shared a multidimensional-stenciled, solution dependent, numerical eigenvalue based method, but essentially differed in the way they recognized the interfaces where the modification of the nonlinear wavespeeds were desired through this dissipation parameter. This led to the HLLC-SWM-E variant (that used an 'Eigenvalue based shock sensor) and the HLLC-SWM-P variant (that used a 'P'ressure based shock sensor). Through a matrix based stability analysis, we obtained a value of α that assured theoretical shock stability over a wide range of free stream Mach numbers.

Further, using a suite of stringent numerical test cases, we showed that both variants of the HLLC-SWM framework provide instability free, clean shock profiles on a variety of problems while, retaining the exact inviscid contact ability of the underlying HLLC scheme. When computing the problem of a subsonic laminar flow over a flat plate, it was found that the HLLC-SWM-P scheme resolved the boundary layer exactly while, the HLLC-SWM-E scheme introduced slight extraneous dissipation into the solution. The success of the HLLC-SWM framework in dealing with shock instability phenomenon alludes to the fact that cures for these spurious failures can be designed by introducing adequate multidimensional dissipation in the vicinity of a shock front. Lastly, the development of the HLLC-SWM-P scheme suggests that it is possible to circumvent the conjecture in Gressier et al. [9] and create a robust shock stable upwind scheme that is accurate on contact and shear waves.

Acknowledgements

The authors would like to thank the reviewer for the useful suggestions during the revision of this paper. The first author would also like to thank Mr. Shainath Kalamkar for proof reading the revised manuscript.

References

- [1] P. Roe, Approximate Riemann solvers, parameter vectors, and difference schemes, *J. Comput. Phys.* 43 (1981) 357–372.
- [2] E.F. Toro, M. Spruce, W. Speares, Restoration of the contact surface in the HLL-Riemann Solver, *Shock Waves* 4 (1994) 25–34.
- [3] B. Einfeldt, C.D. Munz, P.L. Roe, B. Sjogreen, On Godunov-type methods near low densities, *J. Comput. Phys.* 92 (1991) 273–295.
- [4] A. Harten, P.D. Lax, B. van Leer, On upstream differencing and Godunov-type schemes for hyperbolic conservation laws, *SIAM Rev.* 25 (1983) 35–61.
- [5] B. Einfeldt, On Godunov-type methods for gas dynamics, *SIAM J. Numer. Anal.* 25 (2) (1988) 294–318.
- [6] S. Henderson, J. Menart, Grid study on blunt bodies with the carbuncle phenomenon, in: 39th AIAA Thermophysics Conference, Miami, Florida, 2007.
- [7] M. Dumbser, J.M. Moschetta, J. Gressier, A matrix stability analysis of the Carbuncle phenomenon, *J. Comput. Phys.* 197 (2004) 647–670.
- [8] Y. Chauvat, J.M. Moschetta, J. Gressier, Shock wave numerical structure and the Carbuncle phenomenon, *Int. J. Numer. Methods Fluids* 47 (2005) 903–909.
- [9] J. Gressiera, J.-M. Moschetta, Robustness versus accuracy in shock-wave computations, *Int. J. Numer. Methods Fluids* 33 (2000) 313–332.
- [10] J.J. Quirk, A contribution to the great Riemann solver debate, *Int. J. Numer. Methods Fluids* 18 (1994) 555–574.
- [11] M. Pandolfi, D. D'Ambrosio, Numerical instabilities in upwind methods: analysis and cures for the “Carbuncle” phenomenon, *J. Comput. Phys.* 166 (2001) 271–301.
- [12] K. Xu, Z. Li, Dissipative mechanism in Godunov-type schemes, *Int. J. Numer. Methods Fluids* 37 (1) (2001) 1–22.
- [13] W. Xie, W. Li, H. Li, Z. Tian, S. Pan, On numerical instabilities of Godunov-type schemes for strong shocks, *J. Comput. Phys.* 350 (2017) 607–637.
- [14] M.-S. Liou, Mass flux schemes and connection to shock instability, *J. Comput. Phys.* 160 (2000) 623–648.
- [15] Z. Shen, W. Yan, G. Yuan, A robust HLLC-type Riemann solver for strong shock, *J. Comput. Phys.* 309 (2016) 185–206.
- [16] Y.-X. Ren, A robust shock-capturing scheme based on rotated Riemann solvers, *Comput. Fluids* 32 (2003) 1379–1403.
- [17] R. Sanders, E. Morano, M.C. Druguet, Multidimensional dissipation for upwind schemes: stability and applications to gas dynamics, *J. Comput. Phys.* 145 (1998) 511–537.
- [18] Z. Shen, W. Yan, G. Yuan, A stability analysis of Hybrid schemes to cure shock instability, *Commun. Comput. Phys.* 15 (2014) 1320–1342.
- [19] H. Wu, L. Shen, Z. Shen, A hybrid numerical method to cure numerical shock instability, *Commun. Comput. Phys.* 8 (2010) 1264–1271.
- [20] D. Wang, X. Deng, G. Wang, Y. Dong, Developing a hybrid flux function suitable for hypersonic flow simulation with high-order methods, *Int. J. Numer. Methods Fluids* 81 (5) (2016) 309–327.
- [21] H. Nishikawa, K. Kitamura, Very simple carbuncle free boundary layer resolving rotated hybrid Riemann solvers, *J. Comput. Phys.* 227 (2008) 2560–2581.
- [22] K. Huang, H. Wu, H. Yu, D. Yan, Cures for numerical shock instability in HLLC solver, *Int. J. Numer. Methods Fluids* 65 (2011) 1026–1038.
- [23] S. Kim, B. Lee, H. Lee, I.-S. Jeung, Robust HLLC Riemann solver with weighted average flux scheme for strong shock, *J. Comput. Phys.* 228 (2009) 7634–7642.
- [24] S.D. Kim, B.J. Lee, H.J. Lee, I.-S. Jeung, J.-Y. Choi, Realization of contact resolving approximate Riemann solvers for strong shock and expansion flows, *Int. J. Numer. Methods Fluids* 62 (2010) 1107–1133.
- [25] F. Zhang, J. Liu, B. Chen, W. Zhong, Evaluation of rotated upwind schemes for contact discontinuity and strong shock, *Comput. Fluids* 134 (2016) 11–22.
- [26] K.M. Peery, S.T. Imlay, Blunt-Body Flow Simulations, 1988, AIAA Paper 88-2904.
- [27] H.-C. Lin, Dissipation additions to flux-difference splitting, *J. Comput. Phys.* 117 (1) (1995) 20–27.
- [28] A. Harten, J.M. Hyman, Self adjusting grid methods for one-dimensional hyperbolic conservation laws, *J. Comput. Phys.* 50 (2) (1983) 235–269.
- [29] D.S. Balsara, Multidimensional HLLC Riemann solver: application to Euler and magnetohydrodynamic flows, *J. Comput. Phys.* 229 (6) (2010) 1970–1993.
- [30] E. Toro, *Riemann Solvers and Numerical Methods for Fluid Dynamics: A Practical Introduction*, Springer, Berlin, Heidelberg, 2009.
- [31] B. van Leer, J.L. Thomas, P.L. Roe, R.W. Newsome, A Comparison of Numerical Flux Formulas for the Euler and Navier–Stokes Equations, American Institute of Aeronautics and Astronautics Inc, AIAA, 1987, pp. 36–41.
- [32] E. Toro, V. Cendon, Flux splitting schemes for the Euler equations, *Comput. Fluids* 70 (2012) 1–12.
- [33] P. Batten, N. Clarke, C. Lambert, D.M. Causon, On the choice of wavespeeds for the HLLC Riemann solver, *SIAM J. Sci. Comput.* 18 (6) (1997) 1553–1570.
- [34] J. Mandal, V. Panwar, Robust HLL-type Riemann solver capable of resolving contact discontinuity, *Comput. Fluids* 63 (2012) 148–164.
- [35] B. Schmidtman, A.R. Winters, Hybrid entropy stable HLL-type Riemann solvers for hyperbolic conservation laws, *J. Comput. Phys.* 330 (2017) 566–570.
- [36] S.F. Davis, Simplified second-order Godunov-type methods, *SIAM J. Sci. Stat. Comput.* 9 (3) (1988) 445–473.
- [37] S. Simon, J. Mandal, A cure for numerical shock instability in HLLC Riemann solver using antidiffusion control, *Comput. Fluids* 174 (2018) 144–166, <https://doi.org/10.1016/j.compfluid.2018.07.001>.
- [38] F. Zhang, J. Liu, B. Chen, W. Zhong, A robust low-dissipation AUSM-family scheme for numerical shock stability on unstructured grids, *Int. J. Numer. Methods Fluids* 84 (2017) 135–151.
- [39] J. Blazek, *Computational Fluid Dynamics: Principles and Applications*, second edition, Elsevier Science, 2005.
- [40] T. Barth, D. Jespersen, *The Design and Application of Upwind Schemes on Unstructured Meshes*, American Institute of Aeronautics and Astronautics Inc, AIAA, 1989.
- [41] S. Gottlieb, C.-W. Shu, E. Tadmor, Strong stability-preserving high-order time discretization, *SIAM Rev.* 43 (1) (2001) 89–112.
- [42] T. Ohwada, R. Adachi, K. Xu, J. Luo, On the remedy against shock anomalies in kinetic schemes, *J. Comput. Phys.* 255 (2013) 106–129.
- [43] P. Woodward, P. Colella, The numerical simulation of two-dimensional fluid flow with strong shocks, *J. Comput. Phys.* 54 (1) (1984) 115–173.
- [44] Y. Wada, M.-S. Liou, An accurate and robust flux splitting scheme for shock and contact discontinuities, *SIAM J. Sci. Comput.* 18 (1997) 633–657.